WORKED SOLUTION

STEP III 1991, Question 9

Pure • Ellipse Arc Length

STEP III 1991, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q9

The key idea

E_1 is the ellipse in its centre-based form, and E_2 is the same curve in polar form with the pole at a focus. Turning E_2 into Cartesian form and completing the square shows where its centre and axes are, and matching semi-axes gives k and b. For the integrals, the perimeter is one number, but each parametrisation gives a different-looking arc-length integral for it.

Solution

The axes, k and b

Write (x, y) for the coordinates of E_2 and put $r = \sqrt{x^2+y^2}$, $\theta = \theta_2$. Then E_2 says $r = \dfrac{k}{1+e\cos\theta}$, that is $r + e\,r\cos\theta = k$, so

$$r = k - ex.$$

Squaring (this loses nothing, since $r = k/(1+e\cos\theta) > 0$ on the curve),

$$x^2+y^2 = k^2 - 2kex + e^2x^2 \quad\Longrightarrow\quad (1-e^2)x^2 + 2kex + y^2 = k^2.$$

Completing the square in x:

$$(1-e^2)\left(x + \frac{ke}{1-e^2}\right)^2 + y^2 = k^2 + \frac{k^2e^2}{1-e^2} = \frac{k^2}{1-e^2},$$

so

$$\frac{(x+\frac{ke}{1-e^2})^2}{k^2/(1-e^2)^2} + \frac{y^2}{k^2/(1-e^2)} = 1.$$

This is an ellipse with centre $(-\frac{ke}{1-e^2}, 0)$ in the E_2 axes, axes of symmetry parallel to the E_2 axes, and semi-axes

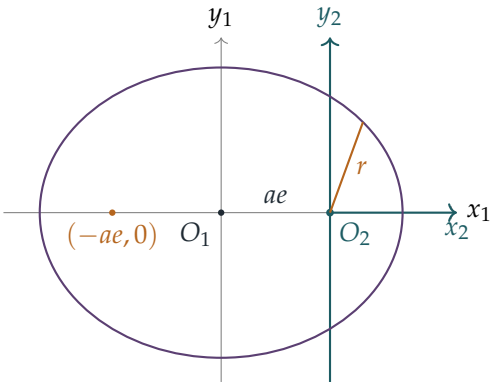
$$A = \frac{k}{1-e^2} \text{ (along the } x\text{-direction)}, \qquad B = \frac{k}{\sqrt{1-e^2}} \text{ (along the } y\text{-direction)}.$$

Since $0 < 1 - e^2 < 1$ we have $A > B$, so the major axis lies along the E_2 x-axis. For E_1, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $a > b$, so its major axis is along the E_1 x-axis. The curves are the same, so the major semi-axes agree and so do the minor ones:

$$a = \frac{k}{1-e^2}, \quad \text{i.e.} \quad k = a(1-e^2), \qquad b^2 = \frac{k^2}{1-e^2} = \frac{a^2(1-e^2)^2}{1-e^2} = a^2(1-e^2),$$

which are the two required results.

The centre, which is the E_1 origin, is at $x = -\frac{ke}{1-e^2} = -ae$ on the E_2 x-axis. So the E_2 axes are the E_1 axes translated a distance ae along the major axis: the E_2 origin is the point $(ae, 0)$ of E_1, which is a focus, since $a^2 - b^2 = a^2e^2$. The two x-axes lie along the same line (the major axis) and the y-axes are parallel; in E_1 coordinates, $x_1 = x_2 + ae$, $y_1 = y_2$. The ellipse is symmetric in both its axes, so we could equally take the E_2 origin at the other focus $(-ae, 0)$ with the x-axis reversed. Either way, the point $\theta_2 = 0$ is the vertex nearest the focus.



The two integrals

We compute the length of the upper half of the ellipse ($y \geq 0$) in two ways.

Using E_1. As θ_1 runs from 0 to π the point runs over the upper half once. With $\dot{x} = -a\sin\theta_1$, $\dot{y} = b\cos\theta_1$,

$$\dot{x}^2 + \dot{y}^2 = a^2\sin^2\theta_1 + b^2\cos^2\theta_1 = a^2 - (a^2 - b^2)\cos^2\theta_1 = a^2\left(1 - e^2\cos^2\theta_1\right),$$

using $a^2 - b^2 = a^2e^2$. So the half-perimeter is

$$\tfrac{1}{2}P = a\int_0^\pi \sqrt{1 - e^2\cos^2\theta}\,\mathrm{d}\theta.$$

Using E_2. Here $y_2 = r\sin\theta_2 \geq 0$ exactly when $0 \leq \theta_2 \leq \pi$, since $r > 0$. So $\theta_2 \in [0, \pi]$ also traces the upper half once. In polar coordinates $\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right)^2 = r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2$. With $r = k(1 + e\cos\theta)^{-1}$,

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{ke\sin\theta}{(1 + e\cos\theta)^2},$$

so

$$r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 = \frac{k^2\left[(1+e\cos\theta)^2 + e^2\sin^2\theta\right]}{(1+e\cos\theta)^4} = \frac{k^2\left(1+e^2+2e\cos\theta\right)}{(1+e\cos\theta)^4}.$$

Hence, using $k = a(1-e^2)$,

$$\tfrac{1}{2}P = a(1-e^2)\int_0^{\pi} \frac{\sqrt{1+e^2+2e\cos\theta}}{(1+e\cos\theta)^2}\,\mathrm{d}\theta.$$

Equating the two expressions for $\frac{1}{2}P$ and dividing by $a > 0$ gives the required identity.

Ask yourself Why must both parameters run over $[0, \pi]$, rather than one of them over a different range?

The arc lengths must be of the same piece of curve. θ_1 and θ_2 are different angles (one is the eccentric angle measured from the centre, the other the polar angle measured from the focus), but each of them, on $[0, \pi]$, sweeps the upper half from the right-hand vertex $(a, 0)$ to the left-hand vertex $(-a, 0)$. So both integrals measure the same arc.

The approximation

Since $0 \le e^2\cos^2\theta < 1$, the binomial series applies:

$$\sqrt{1-u} = 1 - \tfrac{1}{2}u - \tfrac{1}{8}u^2 + O(u^3), \qquad u = e^2\cos^2\theta.$$

The $O(u^3)$ term is $O(e^6)$, which we neglect. Using

$$\int_0^{\pi}\cos^2\theta\,\mathrm{d}\theta = \frac{\pi}{2}, \qquad \int_0^{\pi}\cos^4\theta\,\mathrm{d}\theta = \frac{3\pi}{8}$$

(the second from $\cos^4\theta = \frac{3}{8} + \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta$), we get

$$\begin{aligned}\int_0^{\pi}\sqrt{1-e^2\cos^2\theta}\,\mathrm{d}\theta &\approx \pi - \frac{e^2}{2}\cdot\frac{\pi}{2} - \frac{e^4}{8}\cdot\frac{3\pi}{8} = \pi\left(1 - \frac{e^2}{4} - \frac{3e^4}{64}\right)\\ &= \tfrac{1}{64}\pi\left(64 - 16e^2 - 3e^4\right).\end{aligned}$$

By the identity just proved, the second integral has the same value. That completes the question.

Checks

- At $e = 0$ both integrals equal π, and E_2 becomes the circle $r = k$ with $k = a$: correct.
- With $a = 2$, $b = 1$ (so $e^2 = \frac{3}{4}$), a 200 000-sided polygon gives perimeter 9.6884482, matching $2a$ times the first integral.
- Numerically, at $e = 0.2$ both integrals are 3.1099371 and the approximation gives 3.1099411. The difference is about $-0.063e^6$, consistent with the next term of the series, $-\frac{5\pi}{256}e^6 \approx -0.061e^6$.

Common slips

- Putting the E_2 origin at the centre. The polar form with $1 + e\cos\theta$ in the denominator always has its pole at a focus.
- Forgetting the $(\mathrm{d}r/\mathrm{d}\theta)^2$ term in the polar arc length, or integrating over $[0, 2\pi]$ on one side and $[0, \pi]$ on the other.

Beyond STEP

The integral $\int_0^{\pi/2} \sqrt{1 - e^2\sin^2\theta}\,\mathrm{d}\theta$ is the complete elliptic integral of the second kind, $E(e)$, so the perimeter of an ellipse is $4aE(e)$. It cannot be written in elementary functions, which is why series like the one here (and Ramanujan's remarkably accurate approximations) matter. Elliptic integrals lead on to elliptic functions and elliptic curves, a central topic of modern number theory. For more questions in this style, see Siklos, *Advanced Problems in Mathematics*.

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